NOTRE DAME UNIVERSITY BANGLADESH

Assignment - 01

Course Title: Technical Writing & Presentation Lab

Course Code: CSE 3207

Submitted by:

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Batch: CSE-20

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Assignment Topic: Basic of LaTeX

Submitted to:

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Questions :

1. Write a LaTeX document that demonstrates the following elements from the basics of LaTeX:
   - Create a paragraph that describes LaTeX's purpose, emphasizing the phrase "typesetting system" using the \emph{} command.
   - Use quotation marks to write: "LaTeX is designed for high-quality document preparation."
   - Create a bullet list that includes the following items:
     - Science
     - Technology
     - Engineering
   - Include the following mathematical equation:

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!}$$

2. Write LaTeX code that would help a beginner understand how to create basic text and math formatting:
   - Write a paragraph briefly explaining why LaTeX is useful for creating documents, using both bold (\textbf{}) and italics (\textit{}) to emphasize the key points.
   - Show how to make an enumerated list, listing these steps for LaTeX usage:
     1. Write text with LaTeX commands.
     2. Compile the code.
     3. Check for formatting errors.
   - Insert an example that demonstrates subscript notation using the equation:

$$F_n = F_{n-1} + F_{n-2}$$

   - Write a short note in a comment explaining how to fix common errors when using special symbols like $, %, and & in LaTeX.

Answers :

Code in Texmaker :

```latex
\documentclass[13pt]{article}
\usepackage{amsmath}
\begin{document}
\title{Assignment - 01 : \\ Basic of LaTeX \& Math Formatting}
\author{Istiak Alam}
\maketitle

% Answer to The Question No. 1
\begin{center}
\section*{\underline{Answer to The Question No. 01}}
\end{center}

\subsection*{Basics of LaTeX}
LaTeX is a powerful \emph{typesetting system} designed for the creation of
high-quality documents, particularly those containing complex elements like
mathematical formulas, scientific symbols, and structured formatting. Its
purpose is to provide authors with a consistent and professional layout,
making it ideal for academic papers, theses, research articles, and
technical documentation. Unlike traditional word processors, LaTeX
separates content from formatting, allowing users to focus on writing while
ensuring a clean and precise presentation. Its flexibility and precision
have made it a standard tool in the scientific and academic communities.\\
In short, LaTeX is a document preparation system and a \emph{typesetting
system} widely used for producing scientific and technical documents.\\
\begin{center}
\textbf{``LaTeX is designed for high-quality document preparation.''}
\end{center}

\subsection*{Bullet List Example:}
\begin{itemize}
    \item Science
    \item Technology
    \item Engineering
\end{itemize}

\subsection*{Mathematical Equation:}
\begin{equation*}
    e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots +
\frac{x^n}{n!}
\end{equation*}

% Answer to The Question No. 2
\begin{center}
\section*{\underline{Answer to The Question No. 02}}
\end{center}

\subsection*{LaTeX Code for Text and Math Formatting}
\textbf{LaTeX is useful for creating documents} because it provides
\textbf{high-quality typesetting}, especially for \textit{technical} and
\textit{scientific writing}. It helps maintain \textbf{consistency} and
\textit{professional formatting} throughout large documents. Moreover,
```

```
LaTeX gives users fine control over the structure, citations, equations,
and layout, making it an essential tool in academic and research settings.

\subsection*{Enumerated List for LaTeX Usage}
\begin{enumerate}
    \item \textbf{Write text with LaTeX commands:\\}
    Compose a document using LaTeX syntax for formatting, equations,
sections, etc.
    \item \textbf{Compile the code:\\}
    Use a LaTeX editor or command-line tool (like \texttt{pdflatex}) to
compile the `.tex' file into a PDF.
    \item \textbf{Check for formatting errors:\\}
    Review the output and log messages for any syntax or formatting issues
that need to be corrected.
\end{enumerate}

\subsection*{Subscript Notation}
\textbf{Fibonacci Equation :}
\begin{equation*}
    F_n = F_{n-1} + F_{n-2}
\end{equation*}

% Note for LaTeX beginners:
% Special characters like $, %, &, #, _, {, and } have special meanings in
LaTeX.
% To use them as normal text, you need to escape them using a backslash.
% For example: \$, \%, \&, \_, \{, and \}.

\end{document}
```

Output PDF of Texmaker :

Assignment - 01 :
Basic of LaTeX & Math Formatting

Istiak Alam

April 18, 2025

Answer to The Question No. 01

Basics of LaTeX

LaTeX is a powerful *typesetting system* designed for the creation of high-quality documents, particularly those containing complex elements like mathematical formulas, scientific symbols, and structured formatting. Its purpose is to provide authors with a consistent and professional layout, making it ideal for academic papers, theses, research articles, and technical documentation. Unlike traditional word processors, LaTeX separates content from formatting, allowing users to focus on writing while ensuring a clean and precise presentation. Its flexibility and precision have made it a standard tool in the scientific and academic communities.
In short, LaTeX is a document preparation system and a *typesetting system* widely used for producing scientific and technical documents.

"LaTeX is designed for high-quality document preparation."

Bullet List Example:

- Science
- Technology
- Engineering

Mathematical Equation:

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!}$$

Answer to The Question No. 02

LaTeX Code for Text and Math Formatting

LaTeX is useful for creating documents because it provides **high-quality typesetting**, especially for *technical* and *scientific writing*. It helps maintain **consistency** and *professional formatting* throughout large documents. Moreover, LaTeX gives users fine control over the structure, citations, equations, and layout, making it an essential tool in academic and research settings.

Enumerated List for LaTeX Usage

1. **Write text with LaTeX commands:**
   Compose a document using LaTeX syntax for formatting, equations, sections, etc.

2. **Compile the code:**
   Use a LaTeX editor or command-line tool (like `pdflatex`) to compile the '.tex' file into a PDF.

3. **Check for formatting errors:**
   Review the output and log messages for any syntax or formatting issues that need to be corrected.

Subscript Notation

Fibonacci Equation :

$$F_n = F_{n-1} + F_{n-2}$$